

Histidine-rich glycoprotein binds DNA and RNA and attenuates their capacity to activate the intrinsic coagulation pathway

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Summary

When triggered by factor (F) XII and nucleic acids, we showed that thrombosis in HRG-deficient mice is accelerated compared with that in wild-type mice. In this study, we set out to identify the mechanisms by which nucleic acids promote contact activation, and to determine whether HRG attenuates their effects. DNA or RNA addition to human plasma enhances thrombin generation via the intrinsic pathway and shortens the clotting time. Their effect on the clotting time is seven- to 14-fold greater in HRG-deficient plasma than in control plasma. Investigations into the mechanisms of activation reveal that nucleic acids a) promote FXII activation in the presence of prekallikrein- and high mo-

lecular weight kininogen (HK), and b) enhance thrombin-mediated FXI activation by 10– to 12-fold. Surface plasmon resonance studies show that DNA and RNA bind FXII, FXIIa, HK, FXI, FXIa and thrombin with high affinity. HRG attenuates DNA- and RNA-mediated FXII activation, and FXI activation by FXIIa or by thrombin, suggesting that HRG down regulates the capacity of DNA and RNA to activate the intrinsic pathway. Therefore, HRG attenuates the procoagulant activity of nucleic acids at multiple levels.

Keywords

Coagulation, intrinsic pathway, histidine-rich glycoprotein, DNA, RNA

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Introduction

The intrinsic pathway is initiated by activation of the contact system; the components of which include factor (F) XII, prekallikrein (PK) and high-molecular-weight kininogen (HK) (1). Contact with polyanionic surfaces induces a conformational change in FXII that results in the generation of small amounts of FXIIa; a process known as autoactivation. FXIIa converts PK to kallikrein, which then feeds back to activate more FXII. FXIIa propagates coagulation by activating FXI. FXI and PK circulate in complex with HK, which localises both proteins to polyanionic surfaces and enhances their activation (1). In the presence of such surfaces, thrombin can feed back and activate FXI, thereby promoting its own generation (2). Therefore, polyanionic surfaces are important cofactors in the contact system (2, 3).

The contact system is considered dispensable for haemostasis because patients with congenital deficiencies of FXII, PK or HK do not have a bleeding diathesis; and those with FXI deficiency rarely have spontaneous bleeding and only experience excessive bleeding

with surgery or trauma (4). These observations prompted a revised model of coagulation that diminished the contribution of the contact system to fibrin formation (4–6). However, recent studies prompted a re-examination of the role of the intrinsic pathway in thrombosis. Thus, FXII- or FXI-deficient mice exhibit attenuated thrombosis at sites of arterial or venous injury (7–9). Likewise, inhibition of FXIIa or FXIa with small molecules or antibodies, or knockdown of these factors with antisense oligonucleotides attenuates thrombosis in mice, rats, rabbits and baboons (7, 10–12). Therefore, with mounting evidence for a role of the intrinsic pathway in thrombosis, FXII and FXI have been identified as prime targets for novel antithrombotic strategies.

Emerging evidence that the contact system contributes to the stabilisation and propagation of arterial and venous thrombi has renewed interest in the search for physiological activators of this system (13). These activators include naturally occurring polyphosphates, such as DNA, RNA and inorganic polyphosphate (polyP) (3, 14). DNA and RNA are released from cells that are stimulated, injured, or dying, whereas polyP is released from the

dense granules of activated platelets (3, 14). Neutrophil extracellular traps (NETs), which consist of de-condensed chromatin that is extruded from activated neutrophils, provide another source of DNA (15, 16). Histones, the other major component of chromatin, also promote coagulation, but only in the presence of platelets or red blood cells (17–19). The importance of DNA, RNA and polyP as potential physiological activators of the intrinsic pathway is highlighted by the observation that administration of nucleases, phosphatase, or neutralising polymers that bind these activators attenuates arterial and venous thrombosis in various animal models (3, 15, 16, 20, 21). Although naturally occurring polyphosphates have been identified as potent activators of the contact system, little is known about how activation is regulated.

We have shown that histidine-rich glycoprotein (HRG) modulates the contact system by binding FXIIa with high affinity and attenuating its capacity to undergo autoactivation, or to propagate coagulation by activating FXI (22). HRG is a 75 kDa glycoprotein that circulates in plasma at a concentration of 1.5–2 μ M (23). HRG has a modular domain structure that endows it with the capacity to bind numerous ligands important in coagulation, including fibrin(ogen), DNA, kallikrein and FXIIa (22, 24, 25). Because HRG is released from activated platelets and binds to components of the contact system, it is uniquely positioned to serve as a novel regulator of the intrinsic pathway (22, 24–26). This concept is supported by our recent demonstration that thrombosis is accelerated in HRG-deficient mice, and that HRG repletion protects these mice from thrombosis triggered by FXII and nucleic acids (27). The purpose of this study was to determine the mechanisms by which HRG attenuates contact activation by nucleic acids. Although previous work has shown that activation of the contact system by DNA and RNA is dependent on FXII and PK, little is known about the effects of nucleic acids on FXII autoactivation and FXI activation by FXIIa or thrombin. Accordingly, we first examined the effect of DNA and RNA on these reactions, and then determined whether HRG attenuates their activities.

Materials and methods

Reagents

Human α -thrombin, FXII, FXIIa, FXI, FXIa, PK, HK and corn trypsin inhibitor (CTI) were from Enzyme Research Laboratories (South Bend, IN). Soybean trypsin inhibitor (STI) was from Sigma Aldrich (St. Louis, MO, USA). HRG was isolated from human plasma using a nickel-NTA agarose column as described previously (22) and was labelled with fluorescein isothiocyanate (FITC) according to the manufacturer's specifications (Sigma Aldrich). Dextran sulfate (500 kDa) was from GE Healthcare (Pittsburgh, PA, USA). H-D-Pro-Phe-Arg-*p*-nitroaniline (S-2302) and pyroGlu-Pro-Arg-*p*-nitroaniline (CS-21(66)) were from Chromogenix (Milano, Italy) and Aniara (Neuville-sur-Oise, France), respectively. D-Phe-Pro-Arg-chloromethyl ketone (FPR) was from EMD Millipore (Toronto, ON, Canada) and FPR-adducts of FXIIa, FXIa and thrombin were prepared as described (28). Recombinant hirudin (lepirudin) was from Dade-Behring

(Marburg, Germany). Phosphatidylcholine and phosphatidylserine (PCPS) vesicles were prepared in a 3:1 ratio as described (29). Lyophilised single-stranded (ss)-DNA from calf thymus and double stranded (ds)-DNA from salmon sperm were from Sigma Aldrich and Invitrogen (Carlsbad, CA, USA), respectively.

Human plasma

Blood collected from 10–15 healthy volunteers was used to prepare platelet poor plasma as described (30). Human pooled plasma was rendered deficient of HRG by affinity chromatography using an immobilised HRG-directed antibody (22). A polyclonal antibody raised against human HRG (Affinity Biologicals, Ancaster, ON, Canada) was affinity purified using an HRG-Sepharose column, prepared as described (27). As a control, mock-deficient plasma was prepared by subjecting plasma to chromatography on unmodified Sepharose. FXII-, FXI- and FVII-deficient human plasmas were from Affinity Biologicals Inc.

Nucleic acid preparation

A549 cells, generously provided by Dr. Patricia Liaw, were cultured in Petri dishes containing RPMI-1640 medium (Invitrogen) supplemented with 10% heat-inactivated fetal bovine serum and 5% penicillin and streptomycin (Sigma Aldrich). For RNA preparations, cell suspensions were stored in RNA Protect reagent (Qiagen, Mississauga, ON, Canada). After subjecting the suspensions to centrifugation at $10,000 \times g$ for 10–15 minutes (min) to sediment the cells, DNA and RNA were isolated using DNeasy and RNeasy Plus Mini isolation kits, respectively (Qiagen), and quantified by measuring absorbance at 260 nm. The integrity of the DNA and RNA preparations was assessed by electrophoresis on 1.5% and 2% agarose gels, respectively, under non-denaturing conditions. DNA migrated as a narrow band with an apparent size over 10 kilobase pairs, whereas RNA migrated as two distinct bands of 0.8 and 2 kilobase pairs (data not shown). Biotin (B)-labelled DNA and RNA were generated using the Photoprobe (Long Arm) biotin labeling kit (Vector Laboratories, Burlingame, CA, USA) according to the manufacturer's instructions.

Plasma clotting assays

Clotting assays were performed as previously described (22), with some modifications. Assays were performed in wells of a 96-well plate and all reagents were diluted into a 41.7 mM imidazole buffer, pH 7.0. To wells containing 50 μ l of DNA or RNA in final concentrations ranging from 0–25 μ g/ml, was added 50 μ l of citrated mock-deficient or HRG-deficient human plasma containing 15 μ M PCPS. For HRG repletion experiments, wells contained 25 μ l of DNA or RNA at a final concentration of 30 μ g/ml, and 25 μ l of buffer containing HRG at final concentrations of 0 to 3 μ M. In all assays, clotting was initiated by addition of 50 μ l of CaCl_2 to a final concentration of 8.3 mM, and absorbance was monitored at 405 nm at 37°C using a SpectroMax M3 plate reader (Molecular Devices, Sunnyvale, CA, USA). The clot time was taken as the time to reach

half-maximal absorbance, as determined with instrument software. The concentration of nucleic acids that shortened the clot time by 50% compared with that measured in their absence (EC_{50}) was determined by nonlinear regression analysis using a rectangular hyperbola (Table Curve, v4, Jandel Scientific, San Rafael, CA, USA). Experiments were repeated three times, each in duplicate.

Thrombin generation assays

Thrombin generation was quantified using calibrated automated thrombography in wells of black 96-well plates (Costar, Lowell, MA, USA). To wells containing 10 μ l of a 20 μ g/ml (final concentration) DNA or RNA solution in 20 mM Tris, pH 7.4, was added 40 μ l of control, FXII-, FXI- or FVII-deficient plasma. Thrombin generation was initiated by adding 15 mM $CaCl_2$ and monitored by quantifying the hydrolysis of 1 mM Z-Gly-Gly-Arg-AMC (Bachem, Bubendorf, Switzerland), a fluorescent thrombin-directed substrate, using a SpectraMax M5e plate reader (Molecular De-

vices). Thrombin generation profiles and data were analysed using Technothrombin TGA software (Technoclone, Vienna, Austria). All experiments were performed four times.

Factor activation assays

The activation of 150 nM FXII by 25 μ g/ml DNA or RNA, or by 0.7 μ g/ml dextran sulfate was quantified in the absence or presence of 15 nM PK and/or 37.5 nM HK in 10 mM HEPES, pH 7.4, 150 mM NaCl (HBS). To examine the effect of HRG on these reactions, studies were repeated in the presence of HRG up to 2 μ M. In all cases, reaction mixtures were incubated at 37°C for 30 min prior to addition of STI to 100 μ g/ml to inhibit kallikrein activity. FXIIa generation was then quantified by adding 417 μ M S-2302, a FXIIa-directed substrate, and monitoring its hydrolysis at 405 nm using a SpectroMax plate reader. FXIIa activity was normalised relative to that determined in the absence of a polyanionic surface, PK, HK and HRG.

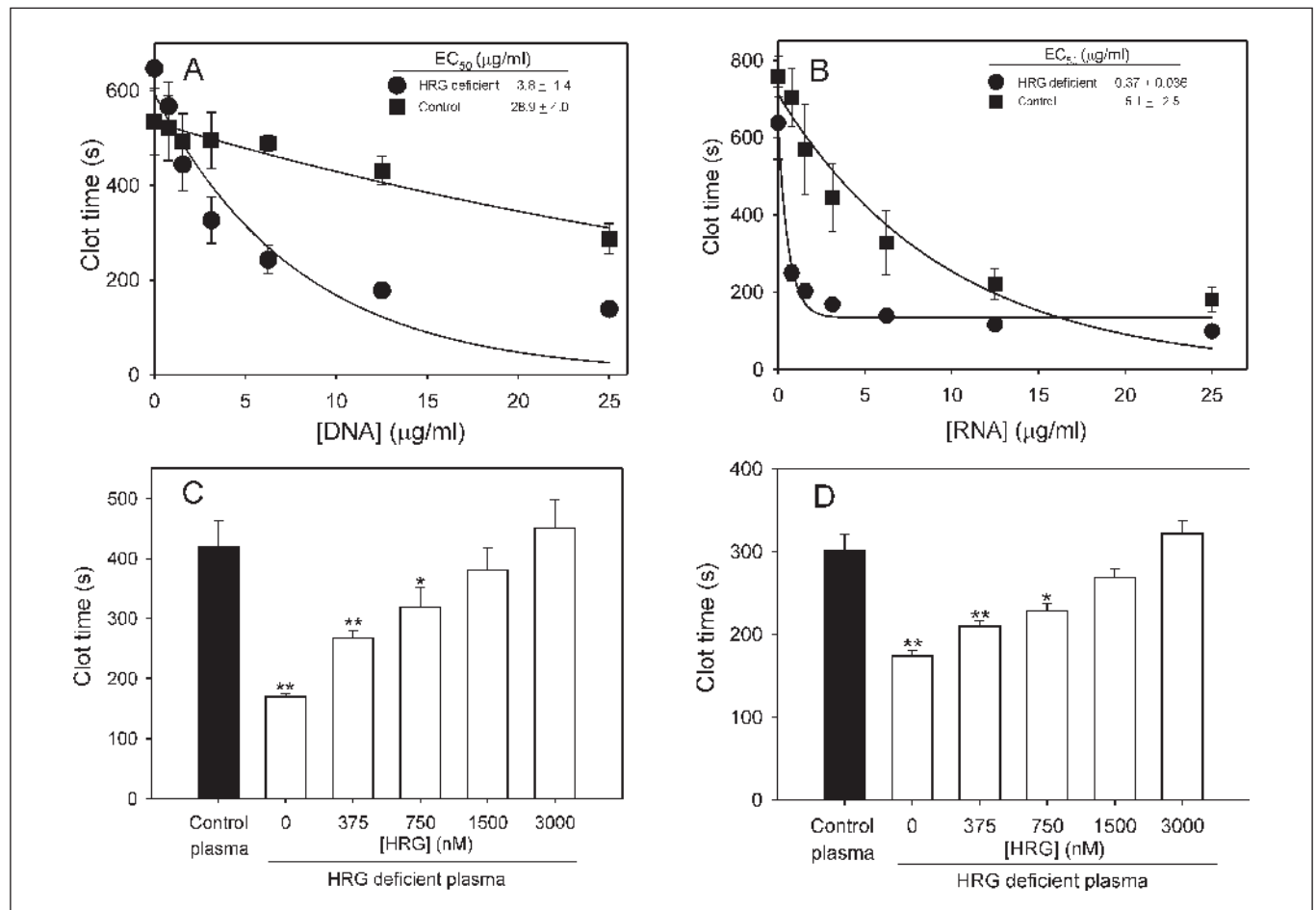


Figure 1: Effect of HRG on DNA- or RNA-induced activation of the intrinsic pathway in plasma. DNA (A) or RNA (B) was added to control (■) or HRG-deficient human plasma (●) and clotting was initiated with $CaCl_2$ to a final concentration of 8.3 mM. Clot times, which were determined as the time to reach half-maximal absorbance at 405 nm, were plotted vs the nucleic acid concentration. EC_{50} values were determined by nonlinear regres-

sion (lines). Clotting was initiated with 30 μ g/ml DNA (C) or RNA (D) in control (black bars) or HRG-deficient human plasma (white bars) supplemented with HRG (0–3 μ M). Data represent the mean $\pm$ SD of three determinations each performed in duplicate. * $p < 0.01$, ** $p < 0.001$, compared with control plasma.

To examine the effect of DNA and RNA on the activation of FXI by FXIIa, 100 nM FXI was incubated with 10 nM FXIIa for 35 min at 37°C in the absence or presence of 20 µg/ml DNA or RNA. After adding 30 nM CTI to inhibit FXIIa, FXIa generation was quantified by monitoring the hydrolysis of 529 µM CS-21(66), a FXIa-directed substrate, at 405 nm. Studies were done in the absence or presence of 100 nM HK and were repeated with or without HRG to 2 µM to determine whether HRG modulates FXI activation by FXIIa. FXIa activity was normalised relative to that determined in the absence of HRG.

To determine whether DNA and RNA promote the activation of FXI by thrombin, 40 nM FXI was incubated with 10 nM thrombin for 35 min at 37°C in the absence or presence of 20 µg/ml of DNA or RNA or 0.7 µg/ml of dextran sulfate. Hirudin was then added to 100 nM to inhibit thrombin activity, and FXIa activity was quantified by monitoring the hydrolysis of 500 µM CS-21(66) at 405 nm. Studies were repeated in the presence of HRG up to 4

µM to determine whether HRG modulates FXI activation by thrombin. FXIa activity in the presence of HRG was normalised relative to that determined in its absence. All experiments were performed three times, and EC₅₀ for HRG was determined by non-linear regression analysis as described above.

Fluorescence measurements

Binding of ss- and ds-DNA to FITC-HRG was assessed by monitoring changes in fluorescence using a Perkin Elmer LS 50B luminescence spectrophotometer (25). Briefly, fluorescence was monitored at excitation and emission wavelengths of 492 and 532 nm, respectively, and slit widths of 10 and 15 nm, respectively, with a 515 nm emission filter. Binding was assessed by titrating aliquots of ss- or ds-DNA in concentrations up to 0.16 µM into a cuvette containing 50 nM FITC-HRG in HBS containing 0.005% Tween 20 (Tw). Intensity values (*I*) after each addition were divided by

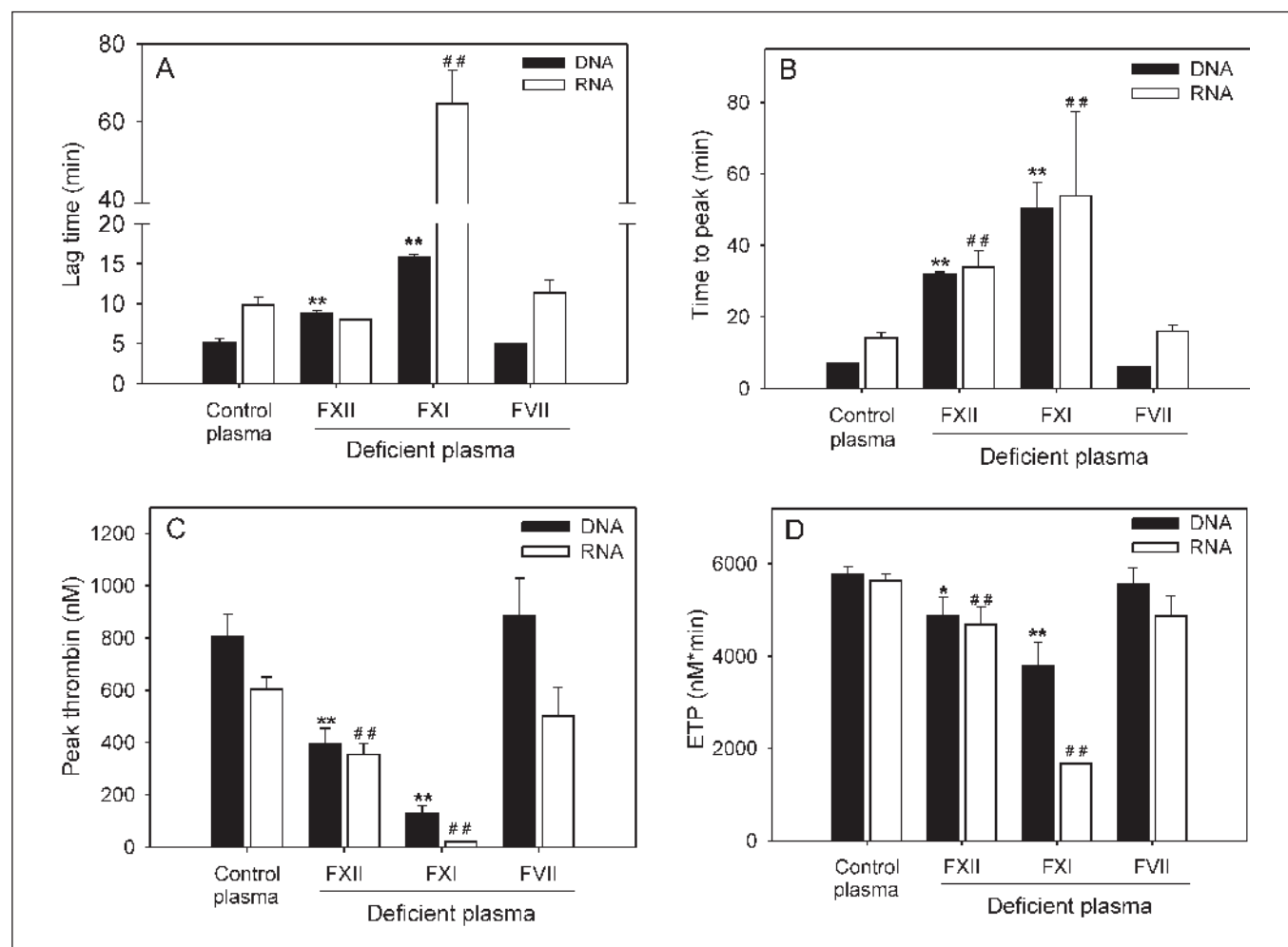


Figure 2: Comparison of DNA- or RNA-induced thrombin generation in control, FXII-, FXI- or FVII-deficient plasma. After incubation of 20 µg/ml DNA (black bars) or RNA (white bars) in control, FXII-, FXI- or FVII-deficient human plasma, 15 mM CaCl₂ was added, thrombin generation was quantified by monitoring the hydrolysis of 1 mM Z-Gly-Gly-Arg-AMC, and

the lag time in min (A), the time to peak thrombin in min (B), the peak thrombin in nM (C), and the ETP in nM*min (D) were determined. Data represent the mean ± SD of four determinations. * *p* < 0.05, ** *p* < 0.005, compared with control plasma in which thrombin generation was initiated with DNA; ## *p* < 0.005, compared with control plasma initiated with RNA.

baseline fluorescence (I_0), plotted against the DNA concentrations, and the data were then subjected to nonlinear regression analysis as described (28). Experiments were performed three times.

Surface plasmon resonance (SPR)

Binding of proteins to immobilised B-labelled DNA or RNA was determined by SPR using a Biacore T200 (GE Healthcare). B-DNA or B-RNA in HBS-Tw was adsorbed to 200–300 response units (RU) onto separate flow cells of a streptavidin-coated chip (XanTec bioanalytics GmbH, Düsseldorf, Germany) at a flow rate of 5 μ l/min. Flow cells were regenerated with 1 M NaCl, 250 mM imidazole and 10 mM EDTA between runs. Aliquots of HRG (0–0.125 μ M), FXII (0–1 μ M), FPR-FXIIa (0–4 μ M), HK (0–2 μ M), PK (0–4 μ M), FXI (0–0.1 μ M), FPR-FXIa (0–0.2 μ M) or FPR-thrombin (0–8 μ M) were injected at a flow rate of 40 μ l/min for 300 seconds (s) to monitor association, followed by injections of HBS-Tw for 800 s to monitor dissociation. An unmodified streptavidin-containing flow cell served as the reference control. Binding of FPR-thrombin and FPR-FXIIa was quantified by steady-state analysis, whereas binding of the other proteins was quantified by global analysis of on- and off-rates using the Langmuir 1:1 binding model, as determined with instrument software (Biacore). All experiments were performed three to four times.

Binding competition

The effect of HRG on DNA or RNA binding to immobilised FXI was assessed by SPR (25). FXI diluted to 50 μ g/ml in 10 mM maleate buffer, pH 6.0 was covalently bound to a flow cell of a CM5 sensor chip in the Biacore T200 using an amine coupling kit (GE Healthcare). To assess association, ds-DNA from salmon sperm or RNA from A549 cells in HBS-Tw at concentrations of 12 μ g/ml and 0.5 μ g/ml, respectively, was injected at a flow rate of 15 μ l/min for 90 s and 120 s, respectively. Dissociation was assessed by addition of HRG (0–5 nM) at a flow rate of 15 μ l/min for 400 s. Between injections, flow cells were regenerated with 1 M NaCl for 240 s. An unmodified flow cell served as the reference control and data were analysed using instrument software. The experiment was repeated three times.

Statistical analysis

Results are presented as the mean $\pm$ SD. Significance of differences was determined using Student t-tests. For all analyses, $p < 0.05$ was considered statistically significant.

Results

HRG attenuates DNA- and RNA-induced activation of coagulation in plasma

DNA or RNA was titrated into control or HRG-deficient human plasma and their effects on clot times were examined. DNA and RNA accelerated clotting in a concentration-dependent fashion in

Table 1: Effect of nucleic acids and contact proteins on FXII activation. The activation of 150 nM FXII by 25 μ g/ml DNA or RNA, or by 0.7 μ g/ml dextran sulfate was quantified in the absence or presence of 15 nM PK and/or 37.5 nM HK and data were normalised relative to the results determined in the absence of DNA, RNA and cofactors. Data represent the mean and standard deviation of three determinations. * $p < 0.05$ and ** $p < 0.005$ vs FXII activation in the absence of polyanion or cofactor, by Student t-test.

	FXIIa generated (relative)			
	No polyanion	DNA	RNA	Dextran sulfate
FXII	1.0	$3.9 \pm 0.6^*$	$4.0 \pm 0.6^*$	52.3 ± 35.1
FXII + HK	1.3 ± 0.03	$3.4 \pm 0.4^*$	3.5 ± 1.6	25.3 ± 15.0
FXII + PK	2.8 ± 1.3	$17.5 \pm 1.4^{**}$	$20.8 \pm 5.3^*$	$209.3 \pm 56.0^*$
FXII + HK + PK	2.2 ± 0.7	$63.0 \pm 8.7^*$	$183.0 \pm 22.1^{**}$	$212.0 \pm 61.6^*$

control and HRG-deficient plasma (► Figure 1A and B, respectively). However, the effective concentration of DNA required to shorten the clot time by 50% (EC_{50}) was seven-fold lower in HRG-deficient plasma than in control plasma (3.8 ± 1.4 and 28.9 ± 4.0 μ g/ml, respectively; $p < 0.01$), whereas the EC_{50} of RNA was 13-fold lower in HRG-deficient plasma than in control plasma (0.4 ± 0.04 and 5.1 ± 2.5 μ g/ml, respectively; $p < 0.01$). Thus, DNA- and RNA-induced clotting in HRG-deficient plasma is enhanced compared with that in control plasma. Addition of HRG to HRG-deficient plasma prolonged the nucleic acid-induced clot times in a concentration-dependent fashion to values similar to those in control plasma (► Figure 1C and D). Therefore, these findings demonstrate that HRG is a potent modulator of nucleic acid-induced activation of coagulation. We next set out to quantify the effects of DNA and RNA on thrombin generation and FXII and FXI activation and to determine how HRG modulates these reactions.

Table 2: Dissociation constants for the interactions of FXII, FPR-FXIIa, HK, PK, FXI, FPR-FXIa and FPR-thrombin with DNA and RNA. The affinities of FXII, FPR-FXIIa, HK, PK, FXI, FPR-FXIa and FPR-thrombin for immobilised DNA or RNA were determined using SPR. The affinities of thrombin and FPR-FXIIa for DNA or RNA were determined by steady state analysis of the plots of Req values vs input thrombin or FPR-FXIIa concentrations. The affinities of FXI, FPR-FXIa (Figure 5), FXII, HK and PK for DNA or RNA were determined by kinetic analysis of the binding sensorgrams. Values represent the mean $\pm$ SD of three to four determinations.

	DNA	RNA
	Kd (nM)	
FXII	206.7 ± 87.6	220.3 ± 108.5
FPR-FXIIa	2963.3 ± 310.5	2766.7 ± 768.2
HK	317.0 ± 60.2	308.9 ± 85.6
PK	No binding	No binding
FXI	3.1 ± 0.4	4.4 ± 0.7
FPR-FXIa	13.6 ± 0.8	15.6 ± 0.2
FPR-thrombin	971.3 ± 211.9	867.7 ± 429.8

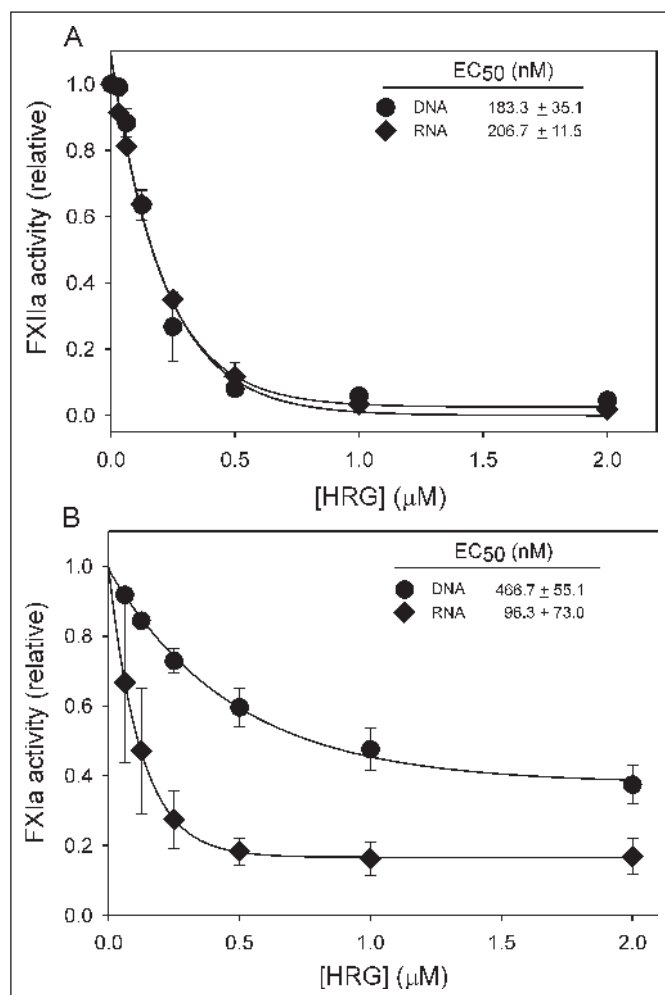


Figure 3: Effect of HRG on DNA- or RNA-induced activation of FXII and FXIIa activation of FXI. A) 150 nM FXII, 15 nM PK, 37.5 nM HK and 25 μg/ml DNA (●) or RNA (◆) were incubated in the presence of 0–2 μM HRG. In the FXIIa assay, kallikrein was inhibited with STI and FXII activation was determined by chromogenic assay. B) For FXI activation, 10 nM FXIIa, 100 nM FXI, 100 nM HK and 20 μg/ml DNA (●) or RNA (◆) were incubated in the presence of 0–2 μM HRG. After 35 min, FXIIa was inhibited with 30nM CTI and FXI activation was quantified by monitoring the hydrolysis of CS-21(66). FXIIa and FXIa activities were normalised relative to those determined in the absence of HRG, and values were plotted vs the HRG concentrations. EC₅₀ values were determined by nonlinear regression (lines). Symbols represent the mean ± SD of three determinations.

DNA and RNA enhance thrombin generation via the intrinsic pathway

In a thrombin generation assay, addition of DNA or RNA to control plasma significantly ($p < 0.005$) shortened the lag time by four-fold and increased endogenous thrombin potential (ETP) by 1.4-fold (data not shown), thereby confirming that nucleic acids are procoagulant (3, 31). However, the capacity of DNA or RNA to promote thrombin generation was attenuated in FXII- or FXI-deficient plasma, but not in FVII deficient plasma (► Figure 2A–D). When DNA or RNA was added to FXII- or FXI-deficient plasma, the time to

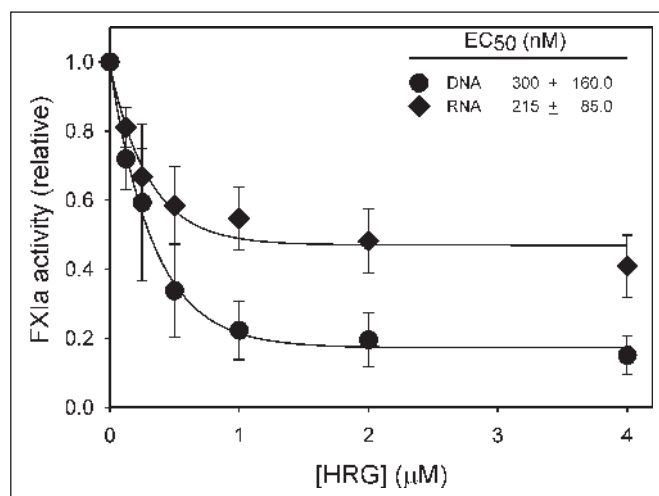


Figure 4: Effect of HRG on thrombin activation of FXI. After incubating 10 nM thrombin and 40 nM FXI with 20 μg/ml DNA (●) or RNA (◆) for 35 min in the absence or presence of 0–4 μM HRG, thrombin was inhibited with 100 nM hirudin and FXI activation was quantified by monitoring the hydrolysis of CS-21(66). FXIa activity in the presence of HRG was normalised relative to that determined in its absence, and values were plotted vs the HRG concentrations. EC₅₀ values were determined by nonlinear regression (lines). Symbols represent the mean ± SD of three determinations.

peak thrombin was significantly prolonged compared with control plasma, and the lag time was significantly prolonged for DNA in both plasmas and for RNA in FXI-deficient plasma (► Figure 2A, B). Similarly, the peak thrombin concentration and ETP in the presence of DNA and RNA were also reduced ($p < 0.005$) in FXII- and FXI-deficient plasma compared with control plasma. Therefore, these data confirm that nucleic acids promote contact activation in a FXII- and FXI-dependent fashion.

DNA and RNA promote FXII activation

We next assessed the capacity of DNA and RNA to activate FXII in the absence or presence of PK and HK. In the absence of HK and PK, DNA and RNA enhanced FXII activation by about four-fold (► Table 1). Whereas HK addition alone had little effect on FXII activation by DNA or RNA, PK addition enhanced activation by up to 20-fold, and addition of both PK and HK promoted FXII activation up to 180-fold (► Table 1). Therefore, although DNA or RNA alone has little effect on FXII activation, these data suggest that they become potent promoters in the presence of PK and HK, in agreement with a previous report (3). By comparison, 0.7 μg/ml dextran sulfate promotes FXII activation by 52-fold even in the absence of PK and HK (► Table 1), suggesting that the mechanism of FXII activation differs depending on the polyanionic activator.

HRG and contact pathway proteins bind to DNA and RNA

Next, we used SPR to quantify binding of contact system proteins to DNA and RNA. FXII and FPR-FXIIa bound to immobilised

nucleic acids with K_d values of ~ 200 nM and ~ 3 μ M, respectively (► Table 2). These data suggest that once FXII is activated, its affinity for nucleic acids decreases. Previous work suggested that HK binds to PK and bridges the zymogen onto polyanionic surfaces (32). Consistent with this concept, we found that HK bound to both DNA and RNA with K_d values of about 300 nM (► Table 2). In contrast, no significant binding was observed with PK. Taken together, these data support the contention that DNA and RNA provide a surface onto which the contact proteins FXII and HK assemble to promote coagulation.

Using SPR, we recently demonstrated that HRG binds DNA and RNA with high affinity, with identical K_d values of 1.3 ± 0.8 nM (27). Binding of HRG to DNA had previously been shown using an immunoassay (24). DNA contains regions that are single-stranded. Because ss-DNA has a structure distinct from that of RNA, we used fluorescence to compare the affinity of FITC-HRG for ds-DNA with that for ss-DNA. FITC-HRG bound ds- and ss-DNA with K_d values of 19.1 ± 1.4 and 28.5 ± 10.6 nM, respectively (data not shown). Therefore, not only are the binding data obtained with fluorescence in agreement with those obtained by SPR, but the fluorescence data reveal that HRG binds ds-DNA and ss-DNA with similar high affinity.

HRG attenuates DNA- or RNA-induced activation of FXII

Having confirmed that nucleic acids promote clotting via the intrinsic pathway, we next set out to identify the reactions that are modulated by HRG. The effect of HRG on DNA- or RNA-induced activation of FXII was examined in the presence of PK and HK because nucleic acids enhanced activation of FXII up to 180-fold in the presence of HK and PK, but only four-fold in their absence (► Table 1). HRG attenuated FXIIa generation in a concentration-dependent fashion producing up to 95% inhibition with an EC_{50} of about 200 nM (► Figure 3A). Therefore, by binding FXIIa, HRG attenuates DNA- or RNA-induced activation of FXII; findings consistent with previous results obtained with dextran sulfate (22).

HRG attenuates DNA- or RNA-induced activation of FXI by FXIIa

In the presence of 20 μ g/ml DNA or RNA, the amount of FXIa generated by incubating FXI with FXIIa was significantly enhanced by two-fold (data not shown); an effect that was attenuated by HRG (► Figure 3B). With 2 μ M HRG, FXI activation by FXIIa in the presence of DNA or RNA was reduced by 63% and 83%, respectively, with EC_{50} values of about 500 and 100 nM, respectively. Therefore, HRG attenuates nucleic acid-mediated FXI activation by FXIIa.

HRG attenuates DNA- or RNA-induced enhancement of FXI activation by thrombin

Thrombin activation of FXI is a key feedback step that is enhanced by polyanionic surfaces such as dextran sulfate and polyP (2, 33).

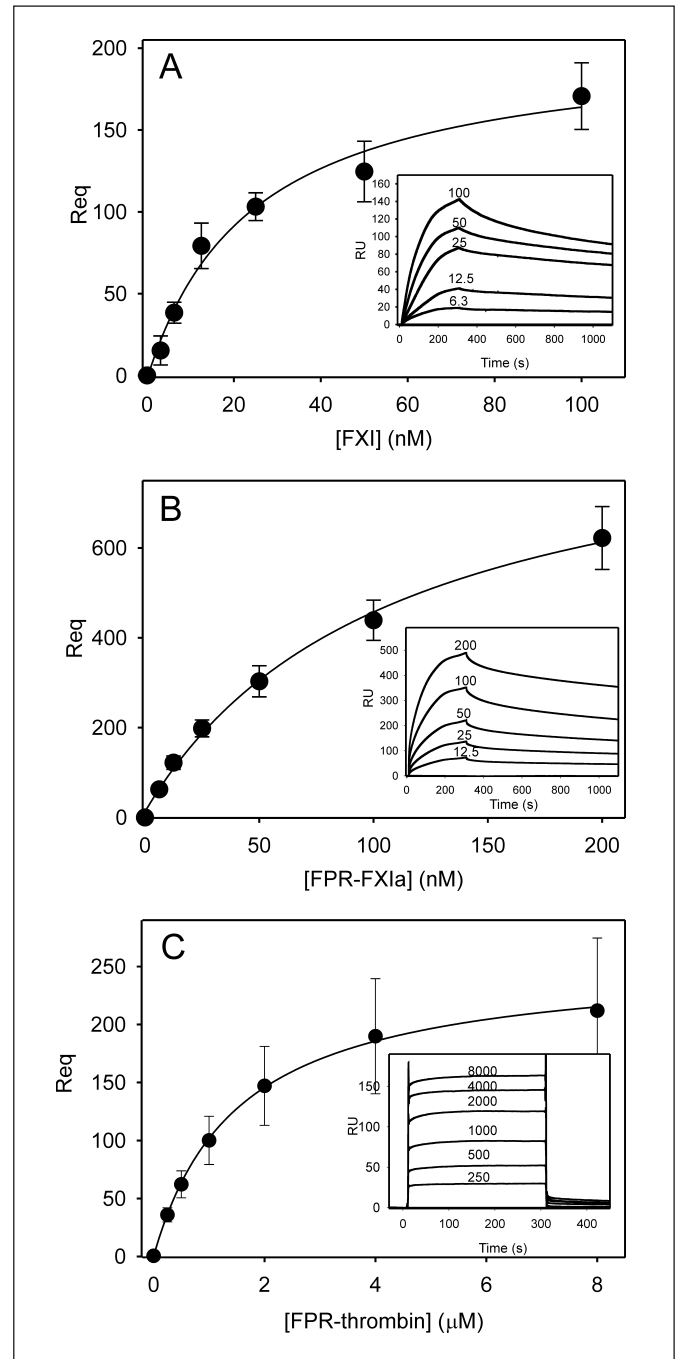


Figure 5: Binding of FXI, FXIa and thrombin to B-DNA as determined using SPR. A) FXI (0–100 nM), B) FPR-FXIa (0–200 nM), or C) FPR-thrombin (0–8 μ M) was separately injected into streptavidin flow cells containing immobilised B-DNA. Req values were plotted vs input protein concentrations. Data represent the mean $\pm$ SD of three determinations and were fit by non-linear regression analysis using a rectangular hyperbola (line). Insets show representative binding sensorgrams from each experiment with B-DNA, where input protein concentrations in nM are indicated adjacent to each tracing.

Previous work has shown that nucleic acids also augment this reaction, so the effect of HRG on this reaction was examined (3). In the presence of DNA or RNA, the extent of thrombin-mediated activation of FXI activation was 10- to 12-fold greater ($p < 0.001$) than that in their absence (data not shown). By comparison, 0.7 $\mu\text{g/ml}$ of dextran sulfate enhanced FXI activation by thrombin by 71-fold ($p < 0.001$; data not shown). HRG attenuated the capacity of DNA or RNA to promote FXI activation by thrombin (► Figure 4). With DNA and RNA, addition of HRG up to 4 μM reduced thrombin-mediated FXI activation by 85% and 60%, respectively, with EC_{50} values of 300 ± 160 and 215 ± 85 nM, respectively. Because HRG does not bind to FXI, FXIa or thrombin (22, 25), these

data raise the possibility that by binding DNA and RNA with high affinity, HRG attenuates their stimulatory capacity.

FXI, FXIa and thrombin bind DNA and RNA

To determine how nucleic acids promote thrombin activation of FXI, we used SPR to quantify the interaction of thrombin, FXI and FXIa with immobilised B-DNA or B-RNA. FXI and FPR-FXIa bound DNA in a concentration-dependent and saturable fashion (► Figure 5A and B, respectively) with K_d values of 3 to 4 nM and 14 to 16 nM, respectively (► Table 2). Similar results were observed with RNA. Therefore, FXI and FXIa bind DNA and RNA with affinities comparable with that of HRG.

Immobilised thrombin has previously been shown to bind polyP with a K_d value of 15 nM (2). We used SPR to determine whether thrombin also binds DNA and RNA. FPR-thrombin bound to immobilised DNA and RNA in a concentration-dependent and saturable fashion (► Figure 5C), yielding K_d values of 900 to 1,000 nM (► Table 2). Compared with HRG, FXI and FXIa, the affinity of thrombin for nucleic acids was much weaker (► Table 2). The capacity of FXI and thrombin to bind nucleic acids supports the contention that DNA and RNA promote thrombin activation of FXI by serving as templates onto which thrombin and FXI assemble (► Figure 4).

HRG displaces FXI bound to DNA or RNA

To determine whether HRG modulates thrombin activation of FXI by binding to the nucleic acid template, we used SPR to assess the capacity of HRG to displace DNA or RNA bound to immobilised FXI. The experiment was designed to exploit the facts that DNA and RNA dissociation from FXI is slow, and that HRG does not bind FXI (22). DNA and RNA bound immobilised FXI in a saturable manner (► Figure 6) and subsequent injection of HRG accelerated their dissociation in a concentration-dependent and saturable manner. Therefore, by binding to the nucleic acids, HRG inhibited FXI binding. These results provide evidence for competitive binding of HRG and FXI to nucleic acids, thereby identifying a secondary mechanism through which HRG regulates the contact pathway of coagulation.

Discussion

Emerging evidence identifies DNA and RNA as important activators of FXII-dependent thrombosis (3, 7, 31, 34). Consequently, it is important to better understand the mechanism by which this occurs and how the activity of nucleic acids is regulated. In this study, we show that nucleic acids promote a) thrombin generation in a FXII- and FXI-dependent fashion, b) FXII activation in the presence of HK and PK, and c) feedback activation of FXI by thrombin. Therefore, these data confirm that, like polyP and mast-cell derived heparin, DNA and RNA are potent activators of the intrinsic pathway. They also demonstrate that nucleic acids impact numerous processes in the intrinsic pathway.

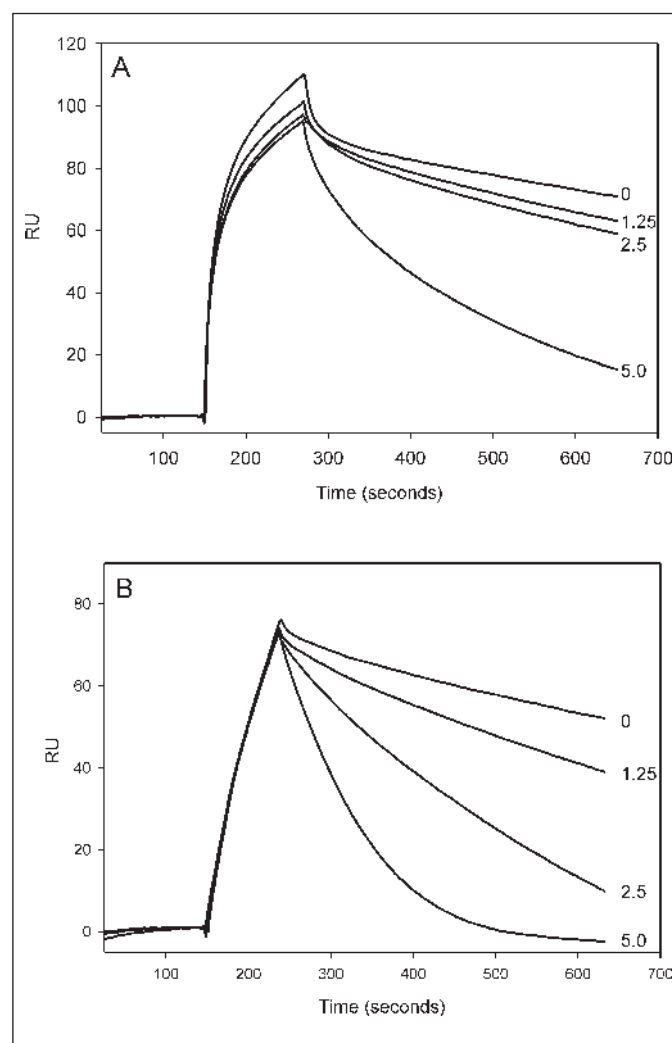


Figure 6: Displacement of DNA and RNA from immobilised FXI by HRG. FXI was immobilised on a flow cell of a CM5 biosensor chip to $\sim 14,000$ RU. A) After injection of 12 $\mu\text{g/ml}$ DNA at a flow rate of 15 $\mu\text{l/min}$ for 120 s, HRG in concentrations from 0–5 nM as indicated was injected at a flow rate of 15 $\mu\text{l/min}$ and dissociation was monitored for 400 s. Between injections, flow cells were regenerated with 1 M NaCl for 240 s. An unmodified flow cell served as the reference control. B) The experiment was then repeated using 0.5 $\mu\text{g/ml}$ RNA in place of DNA. Experiments were repeated three times and representative plots are illustrated.

Apart from degradation by endogenous nucleases, it is unclear how the procoagulant activity of nucleic acids is regulated. Previously, we identified HRG as a regulator of the contact system because it binds FXIIa and attenuates its capacity to promote contact activation of coagulation (22). We subsequently showed that in mice deficient in HRG, thrombosis is accelerated in a nucleic acid-driven arterial injury model, and that administration of HRG reversed this (27). The current study identifies additional mechanisms by which HRG may modulate the procoagulant effects of nucleic acids. Thus, the enhanced activation of FXII by FXIIa and FXI by FXIIa induced by DNA or RNA is attenuated by HRG with EC_{50} values ranging from 100 to 500 nM; concentrations well within the physiological range of HRG of 2 μ M. HRG binds DNA and RNA with K_d values of about 1 nM and attenuates their procoagulant activity because it competes with contact factors for binding to nucleic acids. Therefore, HRG reduces the capacity of DNA and RNA to activate FXII and to support feedback activation of FXI by thrombin. Collectively, our results confirm that nucleic acids are potent activators of the contact system, and indicate that HRG is a dynamic regulator of this pathway.

This study aimed to quantify the stimulatory effects of nucleic acids on individual contact system reactions in order to define the breadth of HRG modulation. The enhanced thrombin generation induced by DNA or RNA is attenuated in FXII- or FXI-deficient plasma, but not in FVII-deficient plasma; findings consistent with the concept that DNA and RNA initiate coagulation exclusively via the intrinsic pathway. Polyanionic surfaces vary in their efficiency as cofactors for FXII activation (35, 36). For example, polyP effectively promotes FXII autoactivation on its own (14), whereas heparin only does so when PK is present (37). Like heparin, DNA and RNA alone are weak activators of FXII, but they promote activation by 180-fold in the presence of PK and HK (► Table 1 and (3). This enhancement likely reflects the reciprocal activation of FXII and PK on the nucleic acid surface. In support of this concept, we show that both FXII and HK bind to DNA and RNA. By bridging PK onto polyanionic surfaces, HK also helps to localize the components of the contact system to the site of activation. Once FXIIa is generated, it propagates the intrinsic pathway by activating FXI; a process amplified by feedback activation of FXI by thrombin (2, 33). Activation of FXI appears to be a pivotal step in nucleic acid-induced activation of the intrinsic pathway because thrombin generation is attenuated to a greater extent in FXI-deficient plasma than in FXII-deficient plasma. Feedback activation of FXI by thrombin requires a polyanionic surface as a cofactor (2, 33). We show that like polyP, DNA and RNA also bind FXI and thrombin and significantly augment the capacity of thrombin to activate FXI; findings consistent with the notion that all polyanions containing an inorganic polyphosphate backbone promote feedback activation of FXI by thrombin (2). Therefore, our collective results suggest that DNA and RNA are cofactors for the activation of FXII and FXI and that these steps are important for initiating, propagating and amplifying the procoagulant response.

Although physiological activators of the intrinsic pathway have now been identified, it remains unclear how this pathway is regulated to prevent uncontrolled clotting. C1 inhibitor is proposed to

be the major regulator of FXIIa, but its rate of FXIIa inhibition is relatively slow compared with that of other serpin-protease interactions (38, 39). Furthermore, when bound to polyanions, FXIIa is protected from regulation by C1 inhibitor (40–42). In contrast, FXII loses affinity for DNA and RNA upon activation, which enables FXIIa to dissociate from the nucleic acids where it can be inhibited by HRG. The capacity of HRG to inhibit the activities of FXIIa, DNA, and RNA establishes it as a potentially important regulator of the intrinsic pathway. Thus, HRG attenuates a) FXII activation, b) FXIIa activity, and c) feedback activation of FXI by thrombin. With its high affinity for DNA and RNA, HRG is localised to the site of contact activation, where it can inhibit FXIIa and neutralise the capacity of DNA or RNA to promote contact activation reactions. Further support for the role of HRG comes from the observation that activated platelets release zinc and HRG, and zinc augments the affinity of HRG for FXIIa by 10^3 -fold (22, 43). Therefore, HRG may serve as a buffer for the contact system in normal haemostasis, and the surge in zinc that occurs when platelets are activated at sites of injury provides a dynamic switch that attenuates thrombosis under conditions of vascular injury. In support of this concept, HRG-FXIIa complexes are detected in contact-activated plasma (22), raising the possibility that HRG was the unidentified plasma protein that interfered with the interaction of a FXIIa-directed nanobody with FXIIa (35).

In summary, we show that nucleic acids are potent cofactors for the activation of FXII and FXI. The identification of physiological activators of the intrinsic pathway suggests that complementary mechanisms exist to down regulate this process. By binding DNA and RNA, HRG is localised to the site of contact activation where it is poised to dampen FXIIa and FXIa generation; processes necessary for the initiation and propagation of the intrinsic pathway of coagulation. These findings add further support for the

What is known about this topic?

- Nucleic acids have been shown to accelerate clotting via the contact pathway.
- Although nucleic acids promote factor XII activation, it is unclear whether they affect other reactions in the contact pathway.
- Histidine-rich glycoprotein binds factor XIIa and attenuates its procoagulant activity, and thrombosis after arterial injury is accelerated in histidine-rich glycoprotein-deficient mice.

What does this paper add?

- DNA and RNA promote factor XII activation in the presence of prekallikrein and high-molecular-weight kininogen, and enhance factor XI activation by thrombin.
- DNA and RNA bind factor XII, high-molecular-weight kininogen, factor XI, and thrombin, thereby serving as potential templates.
- Histidine-rich glycoprotein attenuates DNA- and RNA-stimulated activation of factor XII and XI, demonstrating that it modulates the contact pathway and may serve as endogenous regulator of thrombosis.

concept that HRG serves as a dynamic modulator of the intrinsic pathway.

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Conflicts of interest

None declared.

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